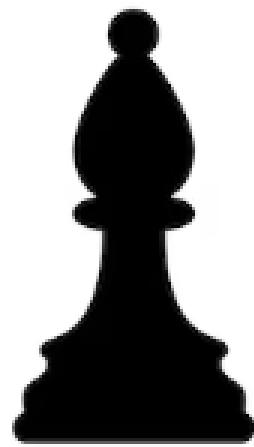


A decorative border made of black and white squares, resembling a chessboard, frames the central text. The border is composed of two horizontal rows of squares at the top and bottom, with vertical columns of squares on the left and right sides, creating a rectangular frame around the text.

CHESS MANIA

TEAM



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ABSTRACT

This report describes the development of a chess AI system capable of evaluating chess positions and predicting optimal moves. The project combines deep learning methodologies with traditional chess evaluation techniques. We employ a Convolutional Neural Network (CNN) architecture that integrates policy and value networks, trained to assess board states and recommend moves. The system is exposed via a FastAPI-based interface, allowing for real-time chess evaluations.

INTRODUCTION



Background on Chess Engines

- **Traditional Engines:** Chess engines like Stockfish and earlier systems rely on brute-force algorithms that evaluate millions of possible positions per second.
- **Search Algorithms:** Engines use minimax and alpha-beta pruning to find the best move.
- **Static Evaluation:** Focused on material count and simple heuristics (e.g., king safety, pawn structure).



Challenges with Traditional Chess Engines

1. Computation-Heavy: Requires significant processing power for deeper search depths.
2. Limited Strategic Understanding: Focuses on immediate tactical gains rather than long-term strategy.
3. No Learning Ability: Static evaluation functions without improvement from previous games.
4. Real-Time Limitations: Not suitable for real-time, interactive environments due to computational overhead.

AI in Chess – A Modern Approach

- Deep Learning: Mimics human thinking by recognizing patterns from past games.
- Neural Networks: Can predict moves and evaluate positions without exhaustive search.
- Learning from Data: Models are trained using vast datasets of historical chess games to improve positional understanding and generalization.
- Goal: Develop a chess AI that offers real-time feedback, strategic depth, and reduced computational requirements using modern machine learning techniques.

EXISTING PROBLEM

- Computational Load: Traditional engines are resource-heavy.
- Positional Understanding: Lack of generalization for strategic depth.
- Real-Time Limitation: Engines struggle to provide instant feedback.
- Hand-Crafted Rules: Rigid evaluation based on static rules.
- No Learning: Traditional engines don't learn or adapt from games.

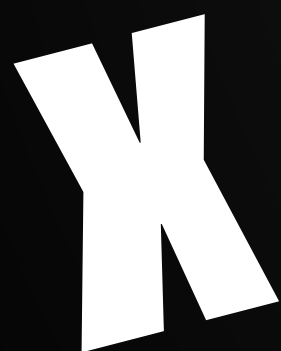




PROPOSED SOLUTION

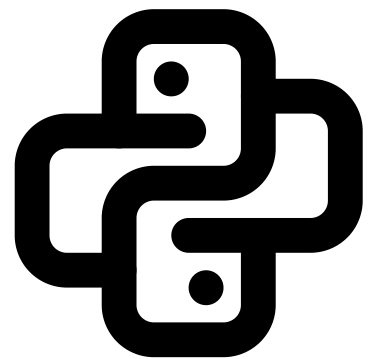


- Neural Network Model: Convolutional Neural Network (CNN) for move prediction (Policy Network) and board evaluation (Value Network).
- Human-Like Intuition: Learns from a vast dataset of games to generalize patterns.
- Real-Time Analysis: FastAPI for instant response to FEN positions.
- Hybrid Approach: Combines deep learning with limited search to refine tactical accuracy.



TEHNOLOGIES USED

- Python & Jupyter Notebook: For development and training.
- TensorFlow: For building the CNN model.
- FastAPI: For deploying the model as an API.
- Python-chess Library: For handling chess rules and move generation.
- CairoSVG: For rendering chessboard images.

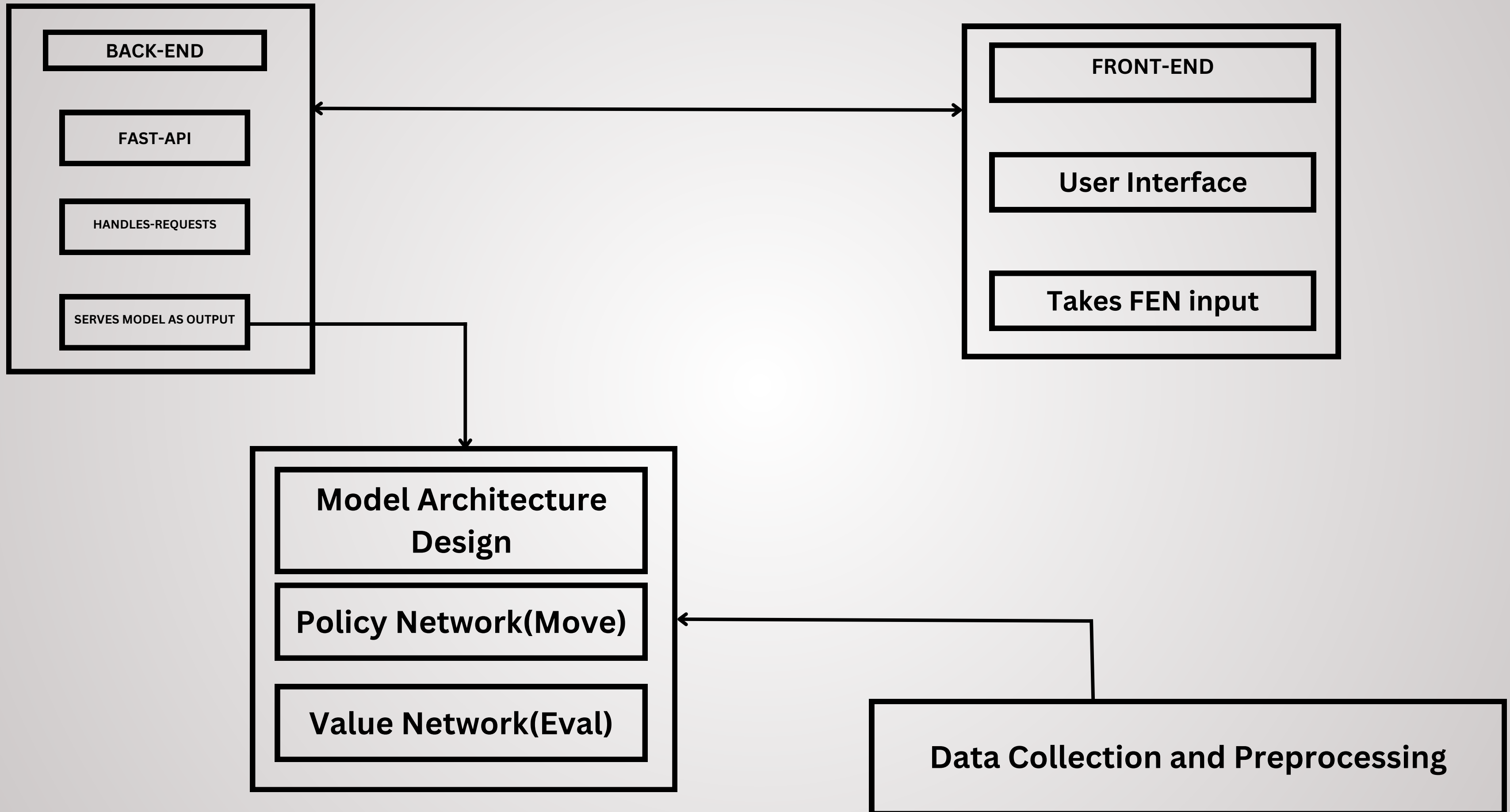


CAIRO
SVG

FAST
API

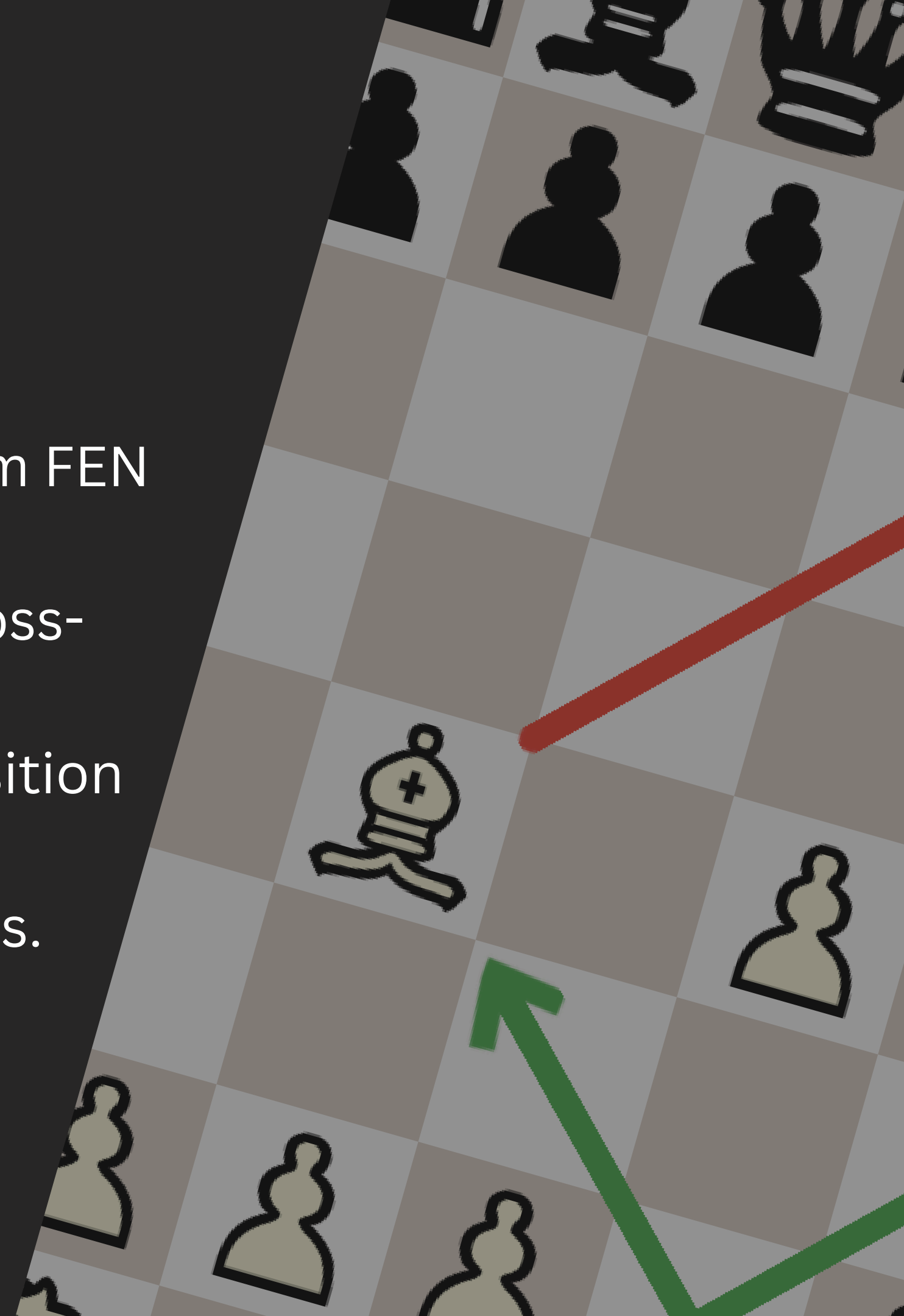
PYTHON
CHESS

ARCHITECTURAL DIAGRAM



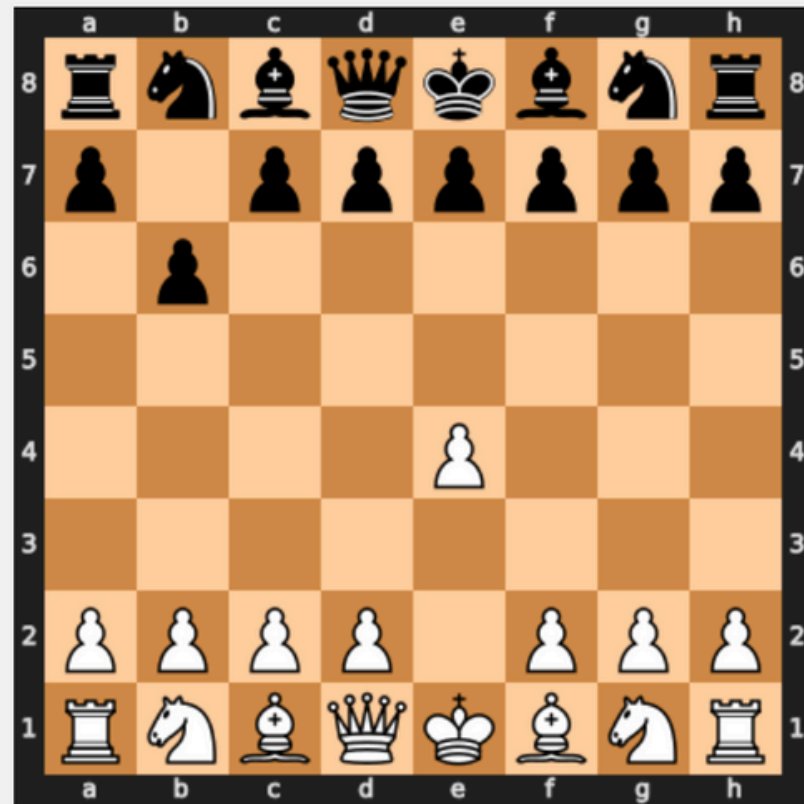
METHODOLOGIES

- Model Architecture: CNN with Policy and Value Networks.
- Data Preprocessing: Convert chess positions from FEN to matrix input.
- Training: Supervised learning with categorical cross-entropy and mean squared error.
- Evaluation: Accuracy of move prediction and position evaluation.
- API Deployment: FastAPI for real-time predictions.



RESULTS

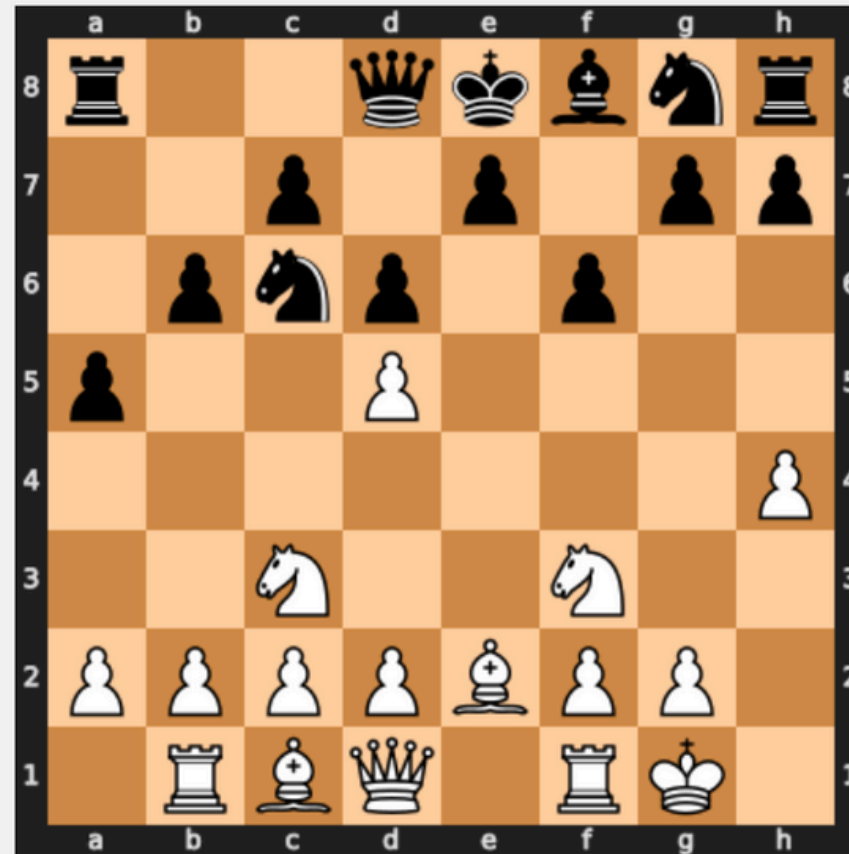
Chess AI Game



Move 4

Play Game

Chess AI Game



Move 16
Random move: a5a4

Play Game

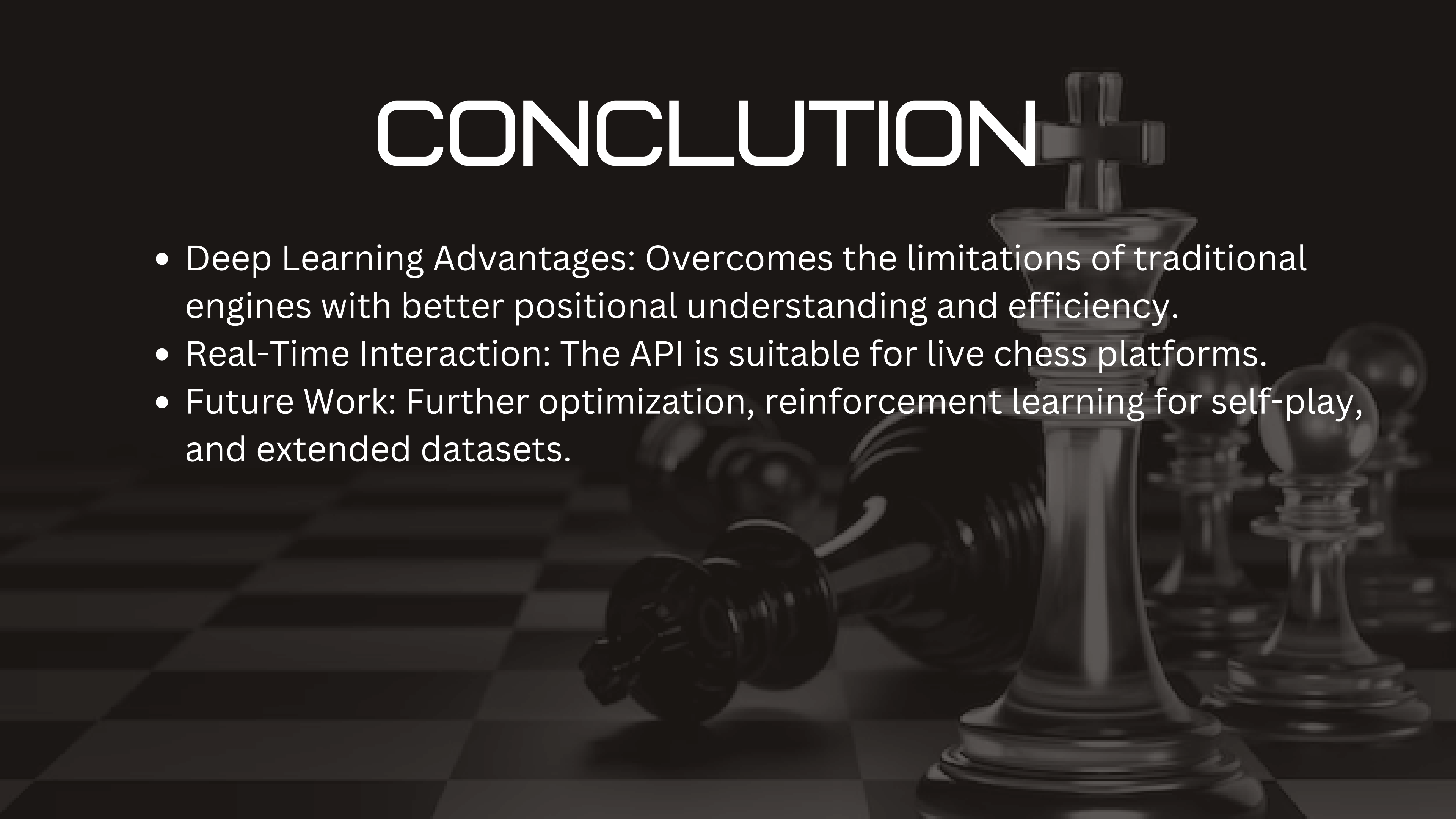
Chess AI Game



Move 25

Play Game

CONCLUSION

A grayscale background image of a chessboard with several chess pieces. A king piece is prominent in the foreground on the right, and a queen piece is visible behind it. To the left, a knight and a rook are also visible. The pieces are on a checkered board, and the lighting creates soft shadows.

- Deep Learning Advantages: Overcomes the limitations of traditional engines with better positional understanding and efficiency.
- Real-Time Interaction: The API is suitable for live chess platforms.
- Future Work: Further optimization, reinforcement learning for self-play, and extended datasets.

REFERENCES

- TensorFlow Documentation: <https://www.tensorflow.org>
- Python-Chess Library: <https://python-chess.readthedocs.io>
- AlphaZero Paper: Silver, D. et al. (2018). "A general reinforcement learning algorithm that masters chess, shogi, and Go through self-play". Science, 362(6419), pp. 1140-1144.
- FastAPI Documentation: <https://fastapi.tiangolo.com>